

Did the Drug Cause It? Automating Causality Assessment

A patient starts a new medicine. Two weeks later their liver fails. Did the drug do it — or the infection they already had, or the other four drugs they take, or nothing to do with any of it? Someone has to decide, case by case, for millions of reports a year. That judgment drives whether a warning gets added to a label or a medicine leaves the market. It is done today largely by hand, by clinicians reading narratives, and it is the least automated high-volume task in drug safety.

Your challenge

Read a real case narrative and produce a structured, auditable causality assessment — the rating, the evidence behind each element, and an honest account of what you could not determine.

The established frameworks give you a scaffold to build against — don't reinvent them, beat them or explain where they fail:

- **Naranjo ADR Probability Scale** 10 questions scored from patient data (prior reports, temporal sequence, dechallenge, rechallenge, alternative causes). Totals map to definite (≥ 9), probable (5–8), possible (1–4), doubtful (≤ 0).
- **WHO-UMC system** category judgment (certain, probable, possible, unlikely, unassessable) weighting clinical plausibility over prior literature.

Reference: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3109846/> (both scales, side by side)

Naranjo detail: <https://www.ncbi.nlm.nih.gov/books/NBK548069/>

Data — real narratives, free, reusable

Public adverse event databases will not work here: FDA's FAERS strips case narratives before release, and narrative is exactly what causality needs. Use the published literature instead.

- **PMC Open Access Subset** millions of full-text articles under licenses permitting reuse and text mining. Filter PubMed to the Case Reports publication type and restrict to the OA subset: drug-induced case reports contain the temporal detail, dechallenge/rechallenge, and competing explanations you need.
- **Built-in reference standard** many published ADR case reports state their own Naranjo or WHO-UMC conclusion. Hold those out and you have something to score against.

Subset and access rules: <https://pmc.ncbi.nlm.nih.gov/tools/openftlist> APIs: <https://www.ncbi.nlm.nih.gov/home/develop/api/>

Retrieve only through E-utilities, the BioC API, FTP, or the cloud service. Scraping the PMC website in bulk is prohibited and will get you blocked mid-hackathon.

Four traps that will cost you points

- **Confident wrongness is the failure mode.** General-purpose models produce fluent causality verdicts that diverge from expert assessment on the same case. Published comparisons already show this. Do not trust your own output because it reads well.
- **The data is mostly missing.** In real practice, 4 of the 10 Naranjo items are answered "unknown" over 85% of the time — and they tend to be the decisive ones. A system that only performs on complete cases solves nothing. Handling absent evidence is the problem, not an edge case.
- **Published cases are a biased sample.** Journals publish the interesting ones. Expect "probable" and "definite" to be wildly overrepresented versus real-world case volume.
- **False negatives cost more than false alarms.** Wrongly clearing a drug is the error that harms patients. Grade yourself on missed associations, not just overall accuracy, and say which direction your system errs in.

What judges reward

- Every element of the rating traceable to specific text in the narrative. A score with no evidence trail is unusable in this field.
- Explicit uncertainty — "cannot determine" is a valid and valuable output.
- An honest number on held-out cases, and disagreement analysis: where do you diverge from the published assessment, and who is right?

Scope and ground rules

Work at the drug-class or therapeutic-area level; do not present conclusions about named products. Public, licensed data only — never patient-identifiable information. Nothing built here is a medical device or clinical decision tool; say so in your demo.

Domain questions are welcome — none of this is common knowledge outside the industry. Prepared for the Regeneron prize track by Global Patient Safety.